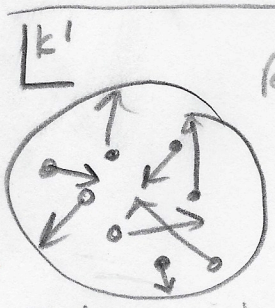
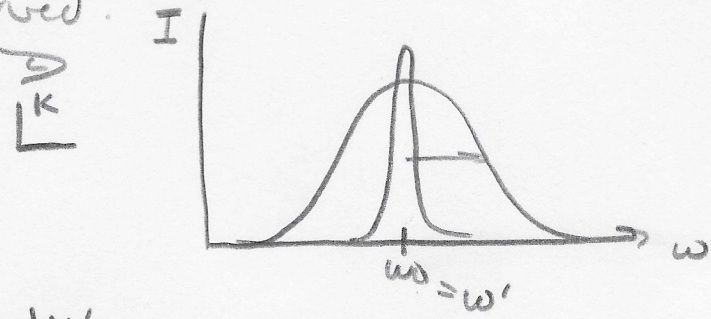
Discussion Notes Thurs. 2/11/16 Doppler Broadening



Random motion of atoms creates shift in light frequency observed.



- Isotropic velocities
- Non-relativistic
- # of photons is invariant; $dN = dN'$.
- # of atoms at a given velocity follows Maxwell-Boltzmann dist.

$$dn \sim e^{-m(v_x^2+v_y^2+v_z^2)/2k_BT}\, dv_x\, dv_y\, dv_z$$

$$\boxed{dn \sim e^{-mv^2/2k_BT}\, v^2\, dv} \quad (1)$$

- Shift in frequency is found with Lorentz transformation of wave vector:

$$\begin{pmatrix} \omega/c \\ k_x \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} \omega'/c \\ k_x' \end{pmatrix} \rightarrow \frac{\omega}{c} = \gamma(\omega' + \beta k_x') = \gamma\left(\frac{\omega'}{c} + \frac{vk\cos\theta'}{c}\right)$$

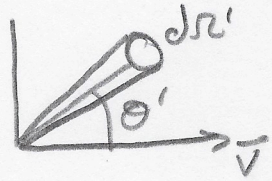
$$\omega = \gamma\omega'(1+\beta\cos\theta')$$

$$\omega = \omega_0(1+\beta\cos\theta')(1-\beta^2)^{-1/2} \simeq \omega_0\left(1+\beta\cos\theta' + \frac{\beta^2}{2} + \frac{\beta^3}{2}\cos\theta'\right) \quad \beta \ll 1$$

$$\boxed{\omega = \omega_0(1+\beta\cos\theta')} \quad (2)$$

- $dN_{TOT} = \int dN\, dn$ ← # of atoms; dN ← # photons per 1 atom $\rightarrow \dfrac{dN_{TOT}}{d\omega} = \int \dfrac{dN}{d\omega}\, dn$

$$\frac{dN'}{d\Omega'} = \frac{dN}{d\Omega'} \rightarrow \frac{dN}{d\omega} \ ?$$



choosing particle rest frame for solid angle since radiation will be isotropic,

$$d\Omega' = \sin\theta' d\theta' d\phi'$$

From (2),

$$\cos\theta' = \frac{1}{\beta}\left(\frac{\omega}{\omega_0} - 1\right)$$

$$-\sin\theta' d\theta' = \frac{d\omega}{\beta\omega_0} \longrightarrow d\Omega' = -\frac{d\omega}{\beta\omega_0} d\phi' = -\frac{2\pi}{\beta\omega_0} d\omega$$

$$\frac{dN}{d\Omega'} = -\frac{\beta\omega_0}{2\pi}\frac{dN}{d\omega} \longrightarrow \frac{dN}{d\omega} = \underbrace{\left(-\frac{2\pi c}{\omega_0}\frac{dN}{d\Omega'}\right)}_{\text{constants} \rightarrow C}\frac{1}{v}$$

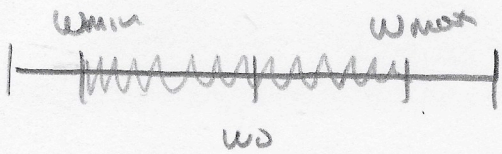
so,

$$\frac{dN_{TOT}}{d\omega} \overset{(1)}{=} C\int \frac{1}{v} e^{-mv^2/2k_BT} v^2\, dv$$ with what limits?

$$\omega = \omega_0\left(1+\frac{v}{c}\cos\theta\right) \rightarrow \omega_{max} = \omega_0\left(1+\frac{v}{c}\cos(0,2\pi)\right) = \omega_0\left(1+\frac{v}{c}\right)$$

$$\omega_{min} = \omega_0\left(1+\frac{v}{c}\cos(\pi)\right) = \omega_0\left(1-\frac{v}{c}\right)$$

There is a range of v's that will produce a range of ω's.



$$\omega_{min} < \omega < \omega_{max}$$

$$\omega_0(1-v/c) < \omega < \omega_0(1+v/c)$$

$$1-\frac{v}{c} < \frac{\omega}{\omega_0} < (1+v/c)$$

$$-v < c\left(\frac{\omega}{\omega_0}-1\right) < v$$

$$-v < c\left(\frac{\omega}{\omega_0}-1\right) \rightarrow v > c\left(1-\frac{\omega}{\omega_0}\right)$$

$$\rightarrow v > c\left(\frac{\omega}{\omega_0}-1\right)$$

or $$v > \left|c\left(\frac{\omega}{\omega_0}-1\right)\right|$$

* This is the lower bound.

* Upper bound for velocity is ∞ because we assume if the particle is moving very fast it must be perpendicular to our detector such that $\cos\theta'$ is zero. Or else we would have been seeing frequencies outside our curve and the analysis would cease to be valid.

$$\frac{dN_{Tot}}{d\omega} = C\int_{|(\frac{\omega}{\omega_0}-1)c|}^{\infty} dv\, v\, e^{-mv^2/2k_BT} = \frac{C}{2}\int_{|(\frac{\omega}{\omega_0}-1)c|^2}^{\infty} dv^2\, e^{-mv^2/2k_BT} = \frac{Ck_BT}{m}e^{-m|(\frac{\omega}{\omega_0}-1)c|^2/2k_BT}$$

$$\boxed{\frac{dN}{d\omega} = \frac{2\pi c k_BT}{m\omega_0}e^{-mc^2(\omega-\omega_0)^2/2\omega_0^2k_BT}}$$

we ignore the (−) from the $d\Omega'$ since it is not physically relevant.